MES AI CONSOLE

Root Cause Analysis AI & Predictive Maintenance

User Manual and AI Implementation Guide

Document Scope

Screen-by-screen operating instructions, AI workflow, model responsibilities, data requirements, governance controls, and expected PI benefits.

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1. Overview

This manual explains the MES (Manufacturing Execution System) AI Console with emphasis on Root Cause Analysis (RCA) and Predictive Maintenance (PM). It also documents the overall menu structure, screen behavior, simulation functions, AI processing logic, and operational considerations.

The console is designed for solar manufacturing processes—INGOT, WAFER, CELL, and MODULE. It uses real-time quality, production, equipment, and sensor data to rank likely defect causes, detect early equipment anomalies, and recommend corrective or preventive actions.

1.1 Primary Functions

- Factory and line dashboards: summarize real-time operating conditions by process and line.
- Root Cause Analysis AI: uses RandomForest feature importance to rank likely contributing factors for a defect.
- Predictive Maintenance AI: uses Isolation Forest-based anomaly detection to identify subtle changes before a conventional threshold alarm.
- LLM explanation: Claude API converts structured model outputs into concise, human-readable explanations and recommended actions.
- Equipment detail: tracks uptime, maintenance history, sensor trends, and estimated Remaining Useful Life (RUL).
- History and reports: filters defect records by process, line, type, and date range.
- Alert and task management: manages detected events through Pending, In Progress, and Done states.

Important

The current package is a demonstration. Displayed AI scores, alerts, emails, SMS messages, and recommendations are simulated unless connected to production data and backend services.

2. Screen Structure and Navigation

The left navigation provides a six-level MES hierarchy. The top status bar displays the system name, live clock, and monitoring status.

Level	Screen	Purpose
1	Factory Overview	Plant-level OEE, alarms, and process status.
2	Line Fleet View	All lines within the selected process.
3	Line Detail · AI Diagnosis	Root Cause Analysis and Predictive Maintenance tabs.
4	Equipment Detail	Uptime, maintenance history, sensor trends, and RUL.
5	History & Reports	Historical defect search and trend visualization.
6	Alert & Task Management	Operational workflow for detected alerts.

The menu number shown on the right side of each navigation item corresponds to its position in the screen hierarchy.

3. Factory Overview

This is the default landing screen after launch. It summarizes the real-time operating condition of the entire plant.

Figure 3-1. Factory Overview screen

3.1 Screen Description

- Use the period tabs to switch among Shift, Daily, Weekly, and Monthly views.
- Process cards summarize key metrics for INGOT, WAFER, CELL, and MODULE.
- Status colors indicate Normal (green), Caution (orange), and Warning (red).
- Selecting a process card opens the Line Fleet View for that process.
- Alarm counts help operators identify areas that require immediate review.

AI Opportunity

A production version can add AI risk indicators, predicted defect trends, and expected downtime impact directly to this overview for faster PI decision-making.

4. Line Fleet View

Select Line Fleet View to compare all lines belonging to a selected manufacturing process.

Figure 4-1. Line Fleet View

4.1 Screen Description

- Select INGOT, WAFER, CELL, or MODULE from the Process dropdown.
- Each line card displays OEE and defect rate.
- Status color is calculated from the current operational metric range.
- Use Add Line to register a new demonstration line.
- Select a line card to open Line Detail · AI Diagnosis.

Operational Use

This view supports rapid prioritization. Lines with low OEE, elevated defect rate, or repeated AI alerts should be reviewed first.

5. Line Detail · AI Diagnosis — Root Cause Analysis

The Line Detail screen contains two tabs: Root Cause Analysis AI and Predictive Maintenance AI. This section explains the RCA tab.

5.1 Process and Line Selection

- Use the top dropdowns to select the current process and line.
- The demonstration scenarios are mapped to representative process lines; CELL uses Line 03.
- Changing the process updates the process schematic, defect scenario, event logs, and probable causes.

5.2 Before Running the Simulation

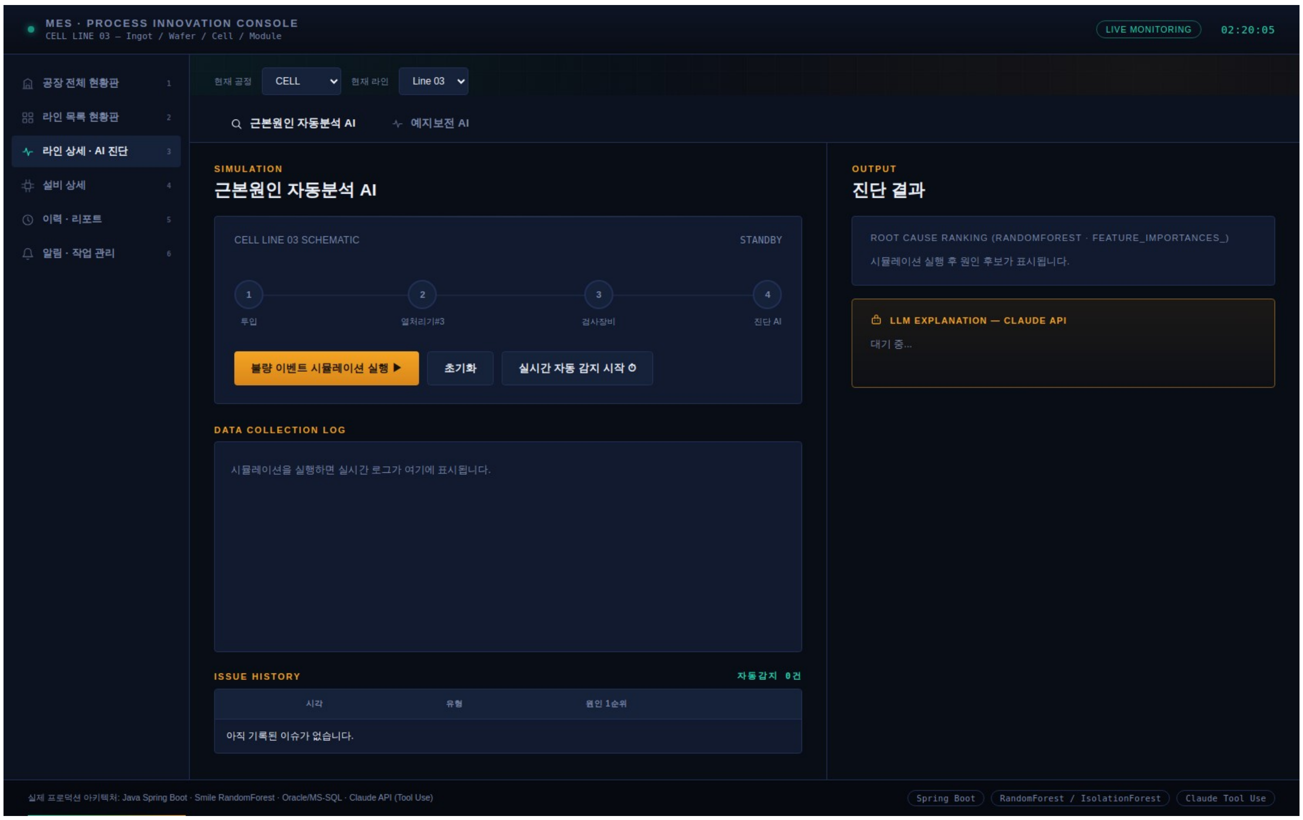

Figure 5-1. Root Cause Analysis — standby state

- The schematic shows the production flow from Input to Heat Treatment, Inspection, and Diagnosis AI.
- Select Simulate Defect Event to replay data collection, model inference, explanation, and notification.
- Select Reset to restore the standby state.
- Select Start Live Auto-Detect to simulate random checks and defect events every 6–11 seconds.
- Data Collection Log records event, fetch, aggregation, model, LLM, and notification steps.
- ISSUE HISTORY accumulates detected issues and provides details when a row is selected.

5.3 After Running the Simulation — Diagnosis Result

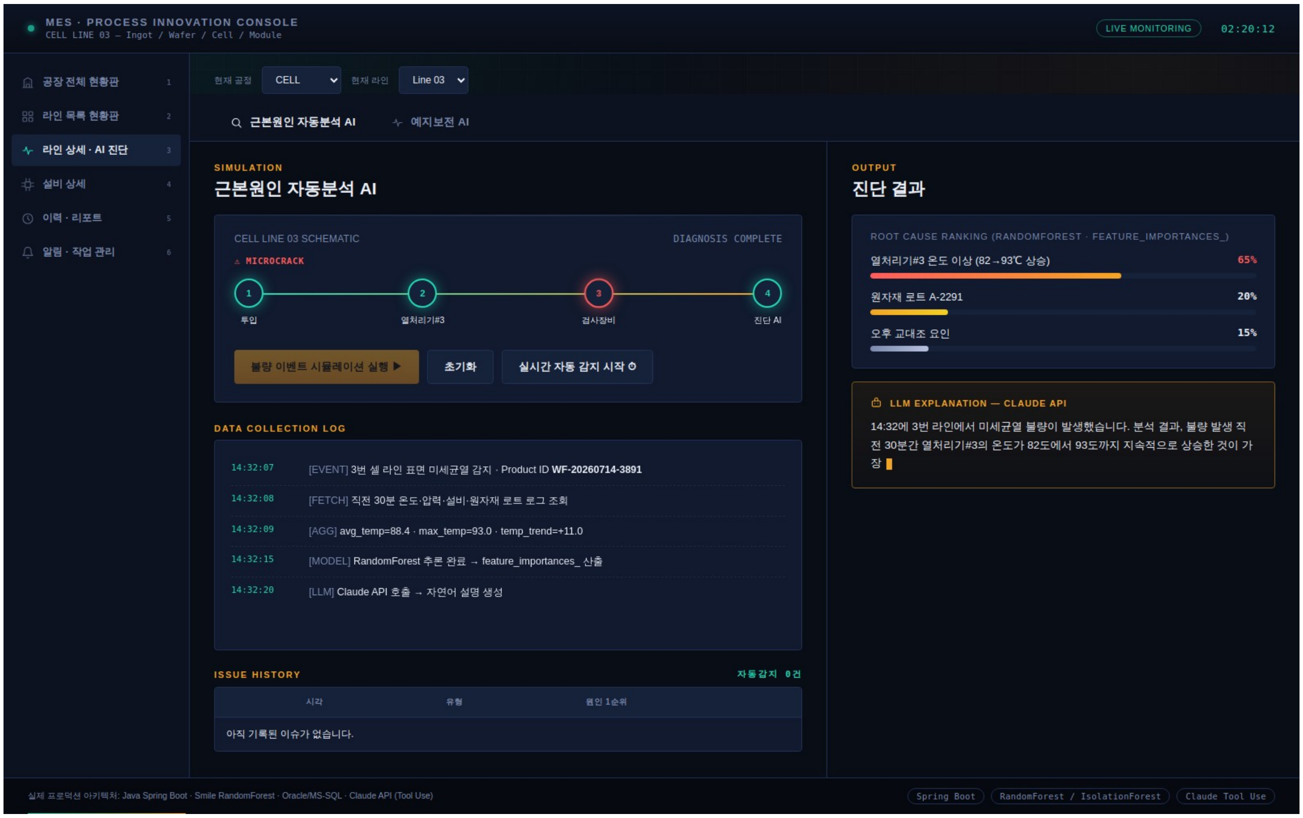

Figure 5-2. Root Cause Analysis — diagnosis completed

- The detected defect type and affected process stage are highlighted.
- Root Cause Ranking lists candidate factors using RandomForest feature importance.
- The displayed percentage represents model contribution or ranking weight in the demo, not guaranteed causal probability.
- LLM Explanation converts the structured results into a concise explanation and recommended first check.
- Notification messages simulate email and SMS delivery to the responsible team.

5.4 Recommended Operator Workflow

1. Confirm that the correct process, line, time window, and defect record were selected.
2. Review the top-ranked contributing factors and underlying sensor or lot data.
3. Compare the result with maintenance history, material traceability, and operating procedures.
4. Perform the recommended inspection or containment action.
5. Record the verified root cause and corrective action for future model improvement.

Human-in-the-Loop Rule

The model ranks likely contributing factors. A qualified engineer must validate the actual root cause before production changes are approved.

6. Line Detail • AI Diagnosis — Predictive Maintenance

Select Predictive Maintenance AI to evaluate equipment condition using time-series sensor data and identify early failure indicators.

6.1 Before Running the Simulation

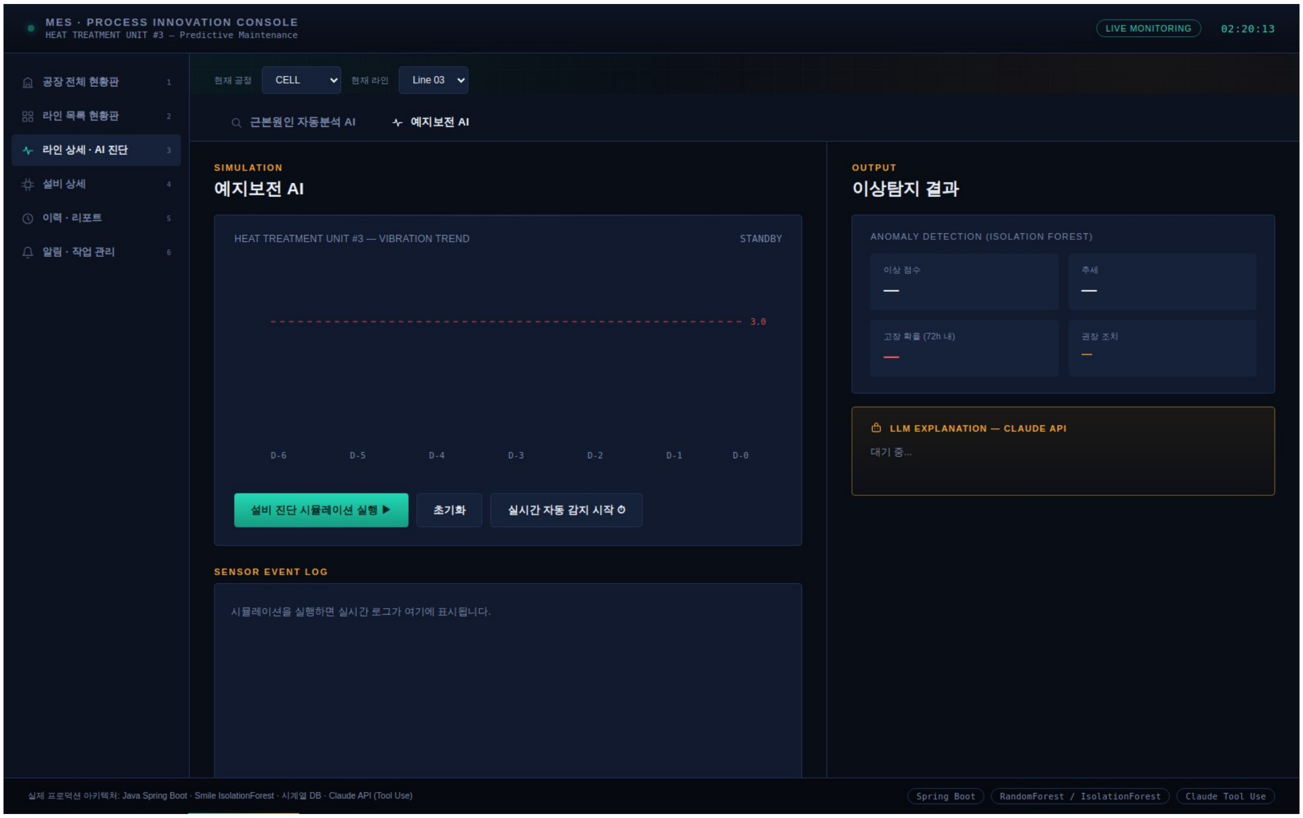

Figure 6-1. Predictive Maintenance — standby state

- The vibration trend chart displays recent equipment behavior from D-6 to D-0.
- The dashed line represents a conventional alarm threshold.
- Select Simulate Equipment Diagnosis to animate the trend and anomaly-detection workflow.
- Reset and Live Auto-Detect operate in the same manner as the RCA tab.

6.2 AI Principle

- Isolation Forest learns the distribution of normal operating patterns and assigns higher anomaly scores to unusual observations.
- Unlike a fixed threshold, anomaly detection can flag multi-variable or gradual changes that still fall within individual normal ranges.
- Failure probability, trend, and recommended action are generated from the demonstration scenario.
- Production deployment should combine vibration, temperature, current, pressure, operating mode, maintenance history, and failure labels where available.

6.3 After Running the Simulation — Anomaly Detection Result

Figure 6-2. Predictive Maintenance — diagnosis completed

- The abnormal region is highlighted and the latest point displays a warning animation.
- The result card summarizes anomaly score, trend, estimated failure probability within 72 hours, and recommended action.
- The LLM explanation describes the detected pattern, expected risk window, and inspection recommendation.
- Email and SMS alerts are simulated after diagnosis completion.

6.4 Recommended Maintenance Workflow

6. Verify sensor quality and confirm the equipment operating mode.
7. Review the anomaly trend across multiple sensors rather than relying on one signal.
8. Check maintenance history and known failure signatures.
9. Schedule inspection during the safest available maintenance window.
10. Record inspection findings as feedback for model validation and retraining.

Key Distinction

Predictive maintenance does not guarantee that a failure will occur. It identifies elevated risk so the team can inspect equipment before an unplanned stop.

7. Equipment Detail

Equipment Detail provides condition and maintenance information for an individual asset.

Figure 7-1. Equipment Detail screen

7.1 Screen Description

- Select a process and equipment item from the dropdowns.
- Review cumulative uptime, last maintenance date, next planned maintenance date, and estimated RUL.
- Switch among 24 Hours, 7 Days, and 30 Days to change the trend view.
- Maintenance History records inspection type, technician, date, and notes.

AI Enhancement

A production version should display model confidence, data freshness, sensor health, baseline version, and the top features that influenced the current equipment risk score.

8. History & Reports

History & Reports supports filtering and trend review of historical defect and diagnosis records.

Figure 8-1. History & Reports screen

8.1 Screen Description

- Filter by process, line, defect type, start date, and end date.
- The trend chart automatically adapts to daily, weekly, or monthly granularity.
- The result table lists timestamp, process, line, defect type, identified cause, and status.
- Report download is simulated in the current demonstration.

8.2 AI Performance Reporting

- Track confirmed versus suggested root causes.
- Measure precision of early-warning alerts and false-alarm rate.
- Measure average lead time between AI alert and actual maintenance finding.
- Compare MTTR, downtime, and defect rate before and after AI adoption.

9. Alert & Task Management

This screen manages alerts produced by RCA and Predictive Maintenance workflows.

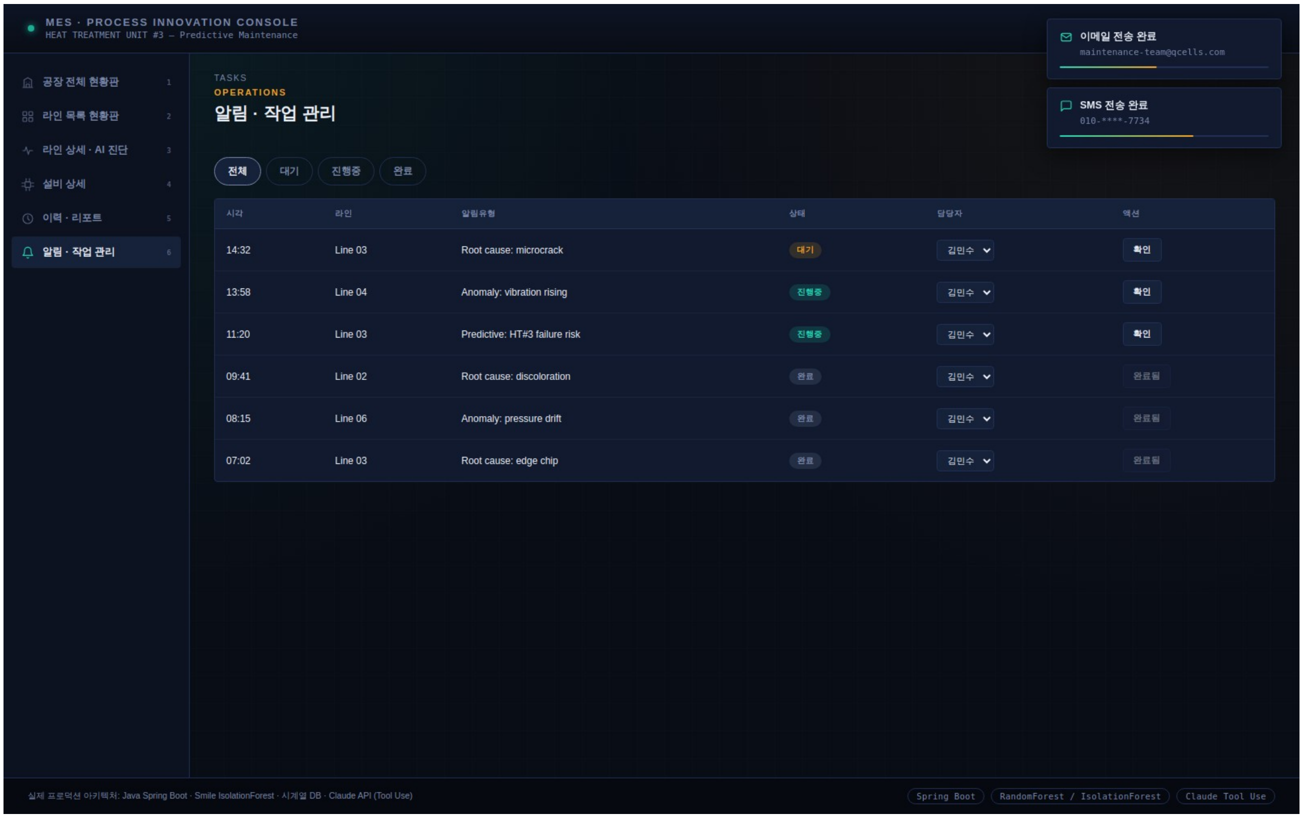

Figure 9-1. Alert & Task Management screen

9.1 Screen Description

- Filter alerts by All, Pending, In Progress, or Done.
- Use the process filter to narrow the list.
- Acknowledge advances an alert from Pending to In Progress and then to Done.
- Each row includes time, process, line, alert type, status, assignee, and action.

9.2 Recommended Alert Controls

- Deduplicate repeated alerts for the same condition.
- Use severity levels based on safety, quality, and production impact.
- Require acknowledgement for critical alerts.
- Escalate unacknowledged alerts based on time and risk.
- Preserve model output, supporting data, user decisions, and final outcome for auditability.

10. AI Architecture and Processing Flow

The recommended production architecture separates data processing, model inference, explanation, and operational decision-making.

Layer	Example Technology	Responsibility	Output
Data Sources	MES / PLC / sensors / quality / ERP	Provide time-aligned process, equipment, quality, lot, and maintenance data.	Validated feature data
Backend	Java / Spring Boot	Collect, validate, aggregate, and expose data through REST APIs.	Feature vector / event context
AI Engine	Smile RandomForest / IsolationForest	Rank contributing factors and detect abnormal patterns.	Scores, rankings, anomaly flags
Database	Oracle / MS SQL	Store events, features, model output, history, and user actions.	Traceable records
LLM Layer	Claude API	Explain structured results and recommended next checks.	Natural-language summary
Frontend	HTML / CSS / JavaScript	Present workflows, logs, results, and task status.	Operator interface

10.1 Root Cause Analysis Flow

11. Detect a defect event and define the relevant time window.
12. Collect process, sensor, equipment, material-lot, quality, and shift data.
13. Validate timestamps, missing values, units, and equipment state.
14. Generate features and run the classification or ranking model.
15. Display top contributing factors with evidence and confidence information.
16. Use the LLM only to explain the structured model output.
17. Require engineer validation before corrective action is finalized.

10.2 Predictive Maintenance Flow

18. Collect real-time time-series data and maintenance context.
19. Create rolling-window features such as mean, standard deviation, slope, frequency features, and rate of change.
20. Run anomaly detection and risk-scoring models.
21. Apply persistence rules to reduce transient false alarms.
22. Generate an alert with supporting trend, risk window, and recommended inspection.
23. Capture the maintenance outcome as model feedback.

11. AI Model Responsibilities and Limitations

11.1 RandomForest for RCA

- Useful for tabular manufacturing data with mixed process, sensor, material, and categorical variables.
- Feature importance helps rank influential variables.
- Feature importance indicates model influence, not proof of causality.
- Correlated variables may share or distort importance.
- Production RCA should combine model ranking with engineering rules, traceability, and causal validation.

11.2 Isolation Forest for Predictive Maintenance

- Useful when failure labels are limited and normal operating data is more abundant.
- Detects unusual combinations of values without requiring a fixed threshold for each sensor.
- Anomaly does not always mean impending failure; changes in recipe, product type, operating mode, or sensor calibration may create alerts.
- Model baselines should be segmented by equipment state and operating condition.

11.3 LLM Explanation

- The LLM should receive structured model outputs, approved terminology, and relevant evidence.
- The LLM should not independently calculate the root cause or failure probability.
- Generated explanations must not introduce unsupported facts.
- Sensitive production data should be minimized, protected, and handled according to company policy.
- High-risk recommendations should use approved templates and require human confirmation.

AI Decision Boundary

AI suggests and explains. Engineers verify, decide, implement, and monitor.

12. Data Requirements, Monitoring, and Governance

12.1 Minimum Data Requirements

- Consistent timestamps across MES, PLC, sensors, inspection, quality, and maintenance systems.
- Equipment identifier, process step, line, product or lot identifier, recipe, and operating mode.
- Defect occurrence time and verified defect classification.
- Sensor values with engineering units, sampling rate, and calibration status.
- Maintenance actions, replaced parts, confirmed failure modes, and inspection findings.

12.2 Model Monitoring

- Data freshness and missing-data rate.
- Feature distribution drift by line, recipe, and equipment state.
- RCA top-cause accuracy after engineer confirmation.
- Predictive-maintenance precision, recall, false-alarm rate, and average warning lead time.
- Model version, threshold version, and explanation template version.

12.3 Governance Controls

- Role-based access to operational and model information.
- Audit logging for model output, explanations, user acknowledgements, and final decisions.
- Version control and approval for models, thresholds, prompts, and recommended actions.
- Fallback behavior when data is stale, missing, or outside the model's validated operating range.
- Periodic review by manufacturing, quality, maintenance, IT, and security stakeholders.

13. PI Benefits and Operational KPIs

The console is intended to support Process Innovation by shortening analysis time, preventing unplanned stops, and standardizing data-driven decisions.

KPI	Meaning	Example Measurement	Target Direction
MTTR	Mean Time to Repair	Time from incident detection to restored operation	Lower
RCA Lead Time	Time to identify the likely cause	Detection to validated root-cause hypothesis	Lower
Unplanned Downtime	Unexpected production stop duration	Hours per line / month	Lower
Alert Precision	Useful alerts among all AI alerts	Confirmed useful alerts ÷ all alerts	Higher
Early Warning Lead Time	Time between alert and required maintenance	Hours or days	Higher, within actionable range
Defect Rate	Defective output proportion	Defects ÷ total production	Lower
Yield	Acceptable output proportion	Good units ÷ total units	Higher
Model Adoption	Operational use of AI output	Acknowledged and reviewed AI cases	Higher

Expected PI Value

Faster RCA, lower MTTR, fewer unplanned stops, more consistent maintenance decisions, improved yield, and traceable operational learning.

14. Appendix — Terms and Acronyms

Term	Full Name / Type	Description
RCA	Root Cause Analysis	Ranks likely contributing factors for a defect or incident.
PM	Predictive Maintenance	Identifies elevated equipment risk before an unplanned failure.
OEE	Overall Equipment Effectiveness	Combined measure of availability, performance, and quality.
RUL	Remaining Useful Life	Estimated time or usage remaining before maintenance or failure.
RandomForest	Ensemble tree model	Used in the demo to rank contributing features.
Isolation Forest	Unsupervised anomaly-detection model	Used to detect unusual operating patterns.
Feature Importance	Model influence measure	Shows which inputs contributed most to the model output; it is not direct causal proof.
LLM	Large Language Model	Converts structured results into natural-language explanations.
MTTR	Mean Time to Repair	Average time required to restore operation after an incident.
Model Drift	Change in data or model behavior	May reduce model reliability over time.
Human-in-the-Loop	Required engineer review	Ensures AI output is validated before operational decisions.
Auto-Detect Mode	Demonstration function	Randomly generates checks and simulated events at set intervals.

Final Note

This manual describes the current demonstration and a recommended production implementation approach. Actual model selection, thresholds, risk scoring, security controls, and operational procedures must be validated with site-specific data and approved by the responsible engineering teams.